

MAMAKAN, Russian FederationWMO: **302520**Lat: **57.8172N**Lon: **114.0160E**Elev: **817**StdP: **14.27**Time Zone: **8.00 (E08)**Period: **06-19**WBAN: **99999****Annual Heating, Humidification, and Ventilation Design Conditions**

Coldest Month	Heating DB		Humidification DP/MCDB and HR						Coldest Month WS/MCDB				MCWS/PCWD to 99.6% DB		WSF
			99.6%			99%			0.4%		1%				
	99.6%	99%	DP	HR	MCDB	DP	HR	MCDB	WS	MCDB	WS	MCDB	MCWS	PCWD	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
1	-46.2	-43.5	-49.4	0.3	-45.9	-46.8	0.4	-43.4	12.4	6.9	10.5	5.5	1.1	80	0.645

Annual Cooling, Dehumidification, and Enthalpy Design Conditions

Hottest Month	Hottest Month DB Range	Cooling DB/MCWB						Evaporation WB/MCDB						MCWS/PCWD to 0.4% DB	
		0.4%		1%		2%		0.4%		1%		2%			
		DB	MCWB	DB	MCWB	DB	MCWB	WB	MCDB	WB	MCDB	WB	MCDB	MCWS	PCWD
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
7	23.3	86.5	68.1	82.5	66.8	78.5	65.3	70.5	81.9	68.4	79.3	66.7	76.3	3.3	90

Dehumidification DP/MCDB and HR									Enthalpy/MCDB						Extreme Max WB
0.4%			1%			2%			0.4%		1%		2%		
DP	HR	MCDB	DP	HR	MCDB	DP	HR	MCDB	Enth	MCDB	Enth	MCDB	Enth	MCDB	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
66.6	101.0	75.5	64.7	94.5	73.8	62.8	88.3	71.3	35.1	82.0	33.3	79.4	31.9	76.9	78.1

Extreme Annual Design Conditions

Extreme Annual WS			Extreme Annual Temperature				n-Year Return Period Values of Extreme Temperature									
			Mean		Standard Deviation		n=5 years		n=10 years		n=20 years		n=50 years			
1%	2.5%	5%	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max		
(a)	(b)	(c)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)		
(4)	11.8	9.5	7.5	DB	-49.9	92.3	3.5	3.3	-52.5	94.7	-54.5	96.6	-56.5	98.5	-59.1	100.9
(5)				WB	-49.7	74.1	3.3	1.8	-52.1	75.4	-54.1	76.5	-56.0	77.5	-58.4	78.8

Monthly Climatic Design Conditions

		Annual	Jan	Feb	Mar	Apr	May	Jun	Jul	Aug	Sep	Oct	Nov	Dec
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
Temperatures, Degree-Days and Degree-Hours	DBAvg	25.1	-20.8	-9.9	10.9	31.9	45.4	61.0	65.7	60.3	45.6	27.7	-0.2	-18.4
	DBStd	32.53	15.56	14.13	14.11	8.94	6.18	7.04	5.44	6.56	7.26	9.84	17.29	15.36
	HDD50	10305	2195	1678	1211	544	172	6	0	9	172	692	1507	2119
	HDD65	14701	2660	2098	1676	994	608	158	58	170	581	1157	1957	2584
	CDD50	1221	0	0	0	0	29	335	487	329	41	0	0	0
	CDD65	142	0	0	0	0	0	36	80	26	0	0	0	0
	CDH74	1942	0	0	0	0	25	676	954	282	5	0	0	0
	CDH80	616	0	0	0	0	3	216	325	72	0	0	0	0
Wind	WSAvg	2.6	1.8	2.2	3.2	3.5	3.6	2.6	2.4	2.3	2.6	3.0	2.2	1.6
Precipitation	PrecAvg	17.8	0.7	0.6	0.4	0.7	1.6	2.6	3.0	3.1	2.1	1.1	1.0	0.8
	PrecMax	22.4	1.3	1.2	0.9	2.0	4.6	4.8	7.5	5.2	3.8	2.6	2.0	1.5
	PrecMin	13.5	0.2	0.2	0.1	0.3	0.7	0.4	0.6	1.1	1.0	0.1	0.4	0.4
	PrecStd	2.4	0.3	0.2	0.2	0.4	0.9	1.1	1.7	1.0	0.7	0.6	0.3	0.3
Monthly Design Dry Bulb and Mean Coincident Wet Bulb Temperatures	0.4%	DB	23.6	30.5	45.9	61.8	77.7	92.2	91.8	87.2	72.6	51.4	43.2	23.5
		MCWB	21.8	27.0	36.7	46.5	57.6	68.2	70.4	69.6	60.0	43.8	37.8	22.1
	2%	DB	13.9	24.7	39.2	56.1	70.0	85.5	87.5	81.0	66.3	46.2	36.1	14.5
		MCWB	12.4	22.5	32.8	42.9	52.3	66.2	68.9	67.9	57.0	40.5	33.2	13.5
	5%	DB	7.3	17.1	34.8	51.2	65.4	81.6	83.6	76.4	61.4	43.0	29.3	8.6
		MCWB	6.4	15.3	30.1	40.7	50.1	64.7	68.1	65.5	54.2	38.9	27.0	7.6
	10%	DB	1.4	11.0	31.5	46.6	60.7	77.4	79.4	72.4	56.9	39.9	24.1	4.0
		MCWB	0.8	9.5	27.4	38.0	47.9	63.0	66.7	63.6	51.8	36.3	22.4	3.5
Monthly Design Wet Bulb and Mean Coincident Dry Bulb Temperatures	0.4%	WB	21.7	27.8	36.9	46.8	58.2	71.6	73.2	71.8	62.5	45.2	38.6	21.6
		MCDB	23.5	30.1	44.9	59.7	74.2	84.6	85.5	83.2	69.9	50.8	42.9	23.5
	2%	WB	12.5	22.3	33.2	44.1	53.9	68.3	71.1	69.1	58.3	41.7	33.2	13.6
		MCDB	13.8	24.9	38.3	54.2	67.0	82.3	82.6	78.7	64.4	45.2	36.0	14.5
	5%	WB	6.5	15.4	30.5	41.2	51.2	66.3	69.3	66.8	55.3	39.4	27.2	7.8
		MCDB	7.2	17.2	34.7	50.5	62.7	78.5	80.1	74.7	60.3	42.7	29.0	8.5
	10%	WB	0.8	9.5	27.6	38.5	48.6	64.2	67.7	64.4	52.3	36.7	22.5	3.5
		MCDB	1.4	10.9	31.5	46.0	59.2	75.2	77.8	70.5	56.2	39.5	24.1	4.0
Mean Daily Temperature Range	5% DB	MDBR	13.3	19.4	26.1	22.3	23.0	26.0	23.3	20.5	17.2	12.9	13.1	11.8
		MCDBR	14.9	20.8	24.0	28.6	34.4	35.2	33.6	29.2	25.6	15.7	13.6	14.1
		MCWBR	13.9	18.2	17.9	16.4	17.9	17.0	16.3	15.8	15.5	10.8	11.0	13.1
	5% WB	MCDBR	15.0	20.7	23.1	26.9	31.1	30.3	28.7	26.1	23.9	14.9	13.6	14.1
		MCWBR	14.0	18.2	17.4	15.9	16.7	15.4	14.6	14.6	14.8	10.6	11.3	13.1
Clear-Sky Solar Irradiance	taub	0.260	0.296	0.320	0.362	0.408	0.442	0.454	0.408	0.346	0.303	0.268	0.244	
	taud	1.997	1.999	2.028	2.038	2.135	2.198	2.199	2.311	2.424	2.323	2.039	1.959	
	Ebn at Noon	177	225	257	266	261	253	246	251	253	229	171	142	
	Edh at Noon	18	30	38	45	44	42	41	35	26	22	17	14	
All-Sky Solar Radiation	RadAvg	216	516	1042	1579	1715	1703	1588	1277	779	422	245	141	
	RadStd	14	32	54	79	130	147	172	113	81	52	30	11	

Historical Trends

	DBAvg	Heating		Cooling			Degree-Days			
		99% DB	99% DP	1% DB	1% WB	1% DP	HDD50	HDD65	CDD50	CDD65
Station Only	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)
Regional (2 neighbors)	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A
	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A